PART 4. ELECTRO MAGNETIC BRAKE AND ITS ADJUSTMENT

1. FUNCTION OF ELECTRO MAGNETIC BRAKE (FIG.3)

; As the electro magnetic brake is operated by spring, it is suitable as safe brake to stop in fuse cutting.

The electro magnetic brake is suitable to vertical movement or hoist system.

(1) RELEASING OF THE BRAKE

: When the magnet(1) is energized, the moving armature(2) is attracted to the magnet(1), and the spring(3) is compressed.

As a result the pressure on the facing(lining)(4) is diminished and the brake is released, the shaft begins to revolve, and starts the hoisting or lowering operation.

(FIG.1 shows the state that the electro magnetic brake is energized and the brake is released.)

(2) BRAKING

: On the contrary when the current is turned off the attractive force of the magnet(1) is lost.

At the same time the spring pushes the armature in the direction of the facing(lining), the armature returnes to the former position.

As a result the braking force is applied to the facing, the revolution of the shaft is stopped, and the hoisting and lowering operation is stopped

2. INSPECTION AND ADJUSTMENT (INSTRUCTION)

; Before beginning the inspection, be sure to disconnect the electric current by turning the power SWITCH OFF.
Otherwise, a serious accident will be caused.
(At the time of adjustment, the case is the same as above.)

(1) The distance of magnet and armature should be 0.5mm on 360°roundly. If they arn't fit, loose the gap fixing bolt(5) and regulate the distance as gap regulation(6) screw.

(2) The electro magnetic brake is operated by facing(lining).
If it is stained with oil or moisture, it lost its function.

(3) Before the trial run, supply the current to the magnet coil only, and confirm that the runing shaft is operating without any resistance.
Then make a trial run after supply the current.

(4) It the slip is occurred in using, the reason is due to the going away of gap.
Then fix the gap to 0.5mm

(5) If you want the more strange torque, regulate screw(6) with closing little by little to clockwise rotation(CW)

ELECTRO MAGNETIC BRAKE

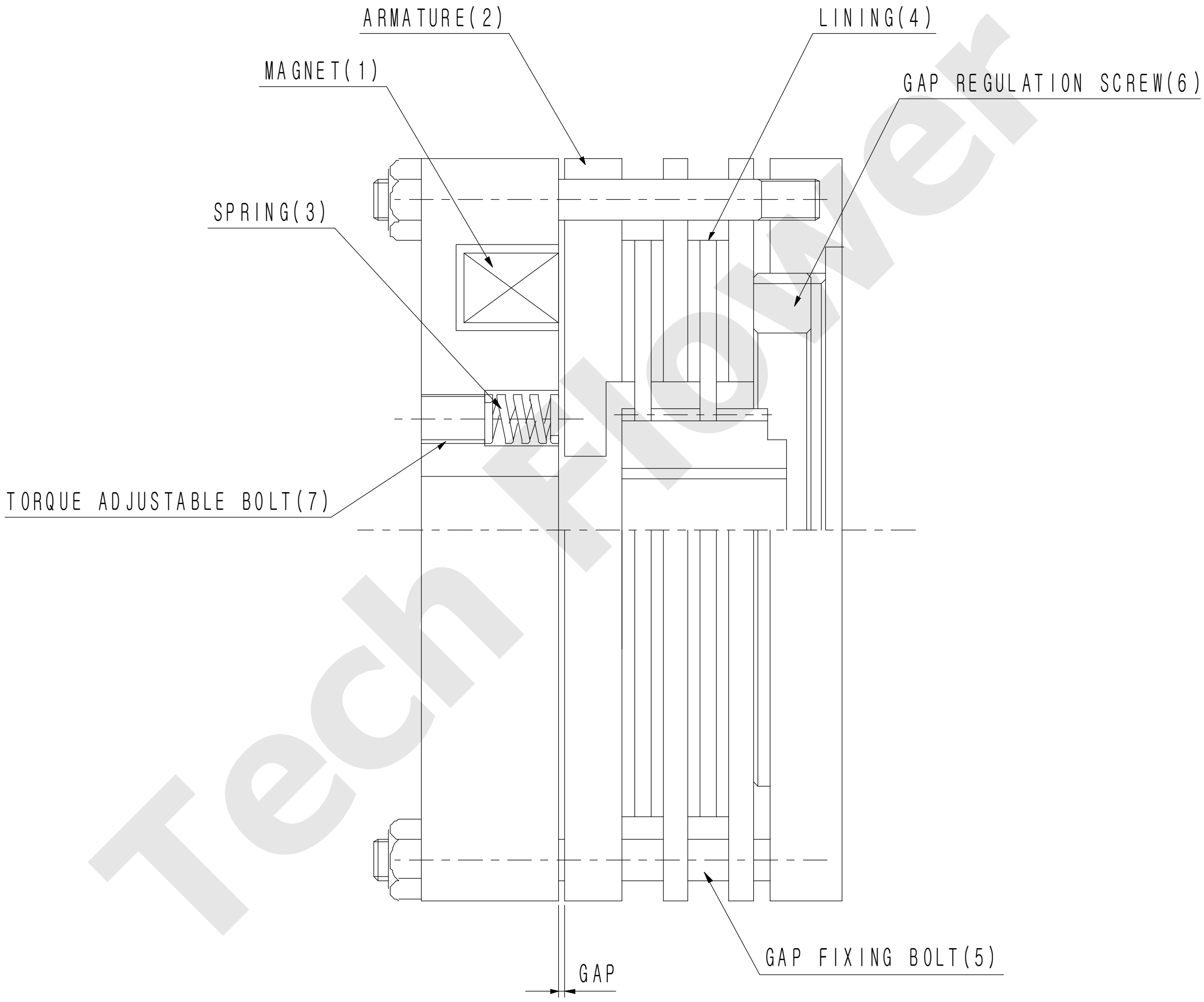

FIG. 3